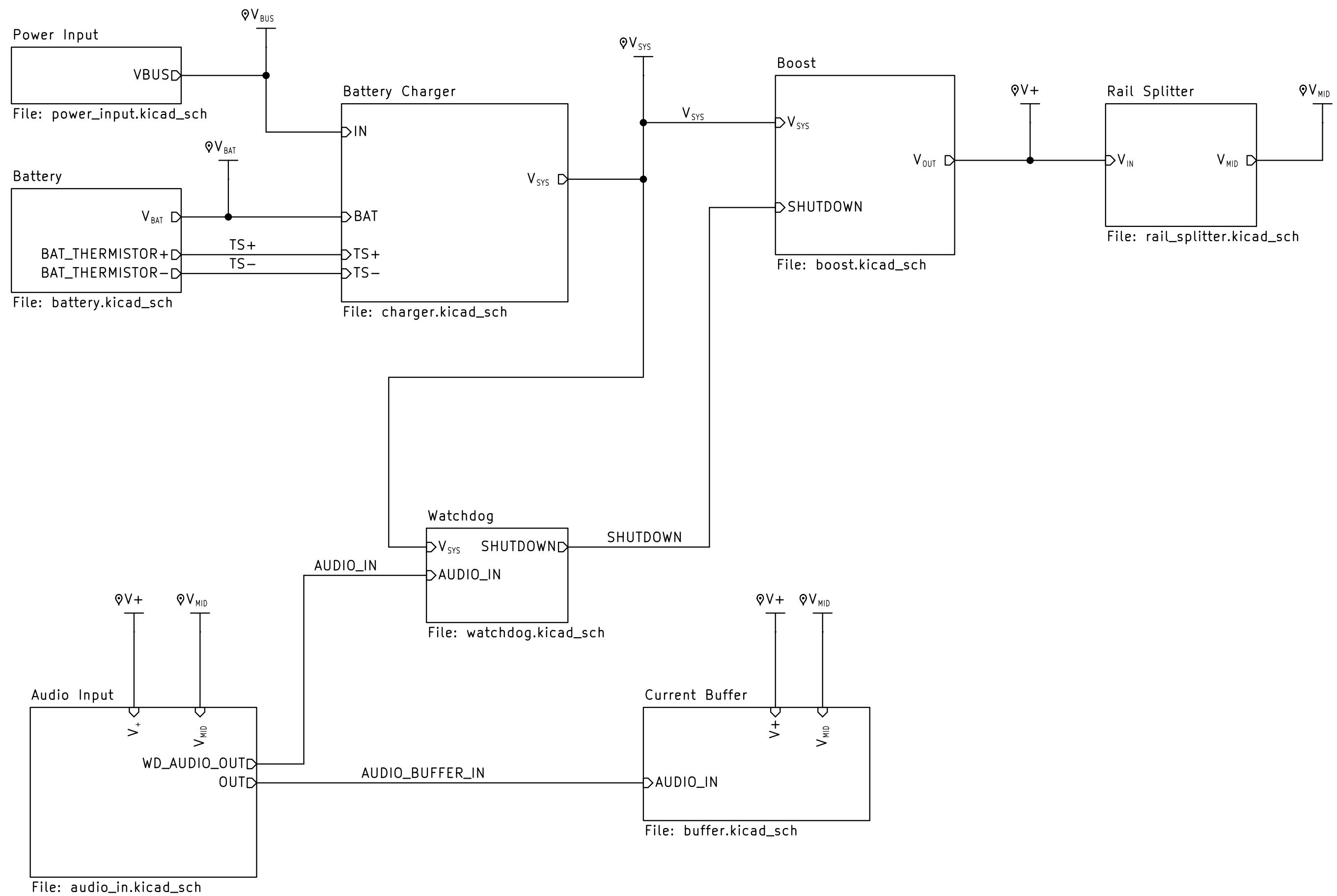


004−0001

Sheet: /
File: headphone_amp.kicad_sch

Title: Portable Headphone Amp

Size: A4 Date: 2026−08−25 **Rev: A1**
KiCad E.D.A. 10.0.0 Id: 1/9

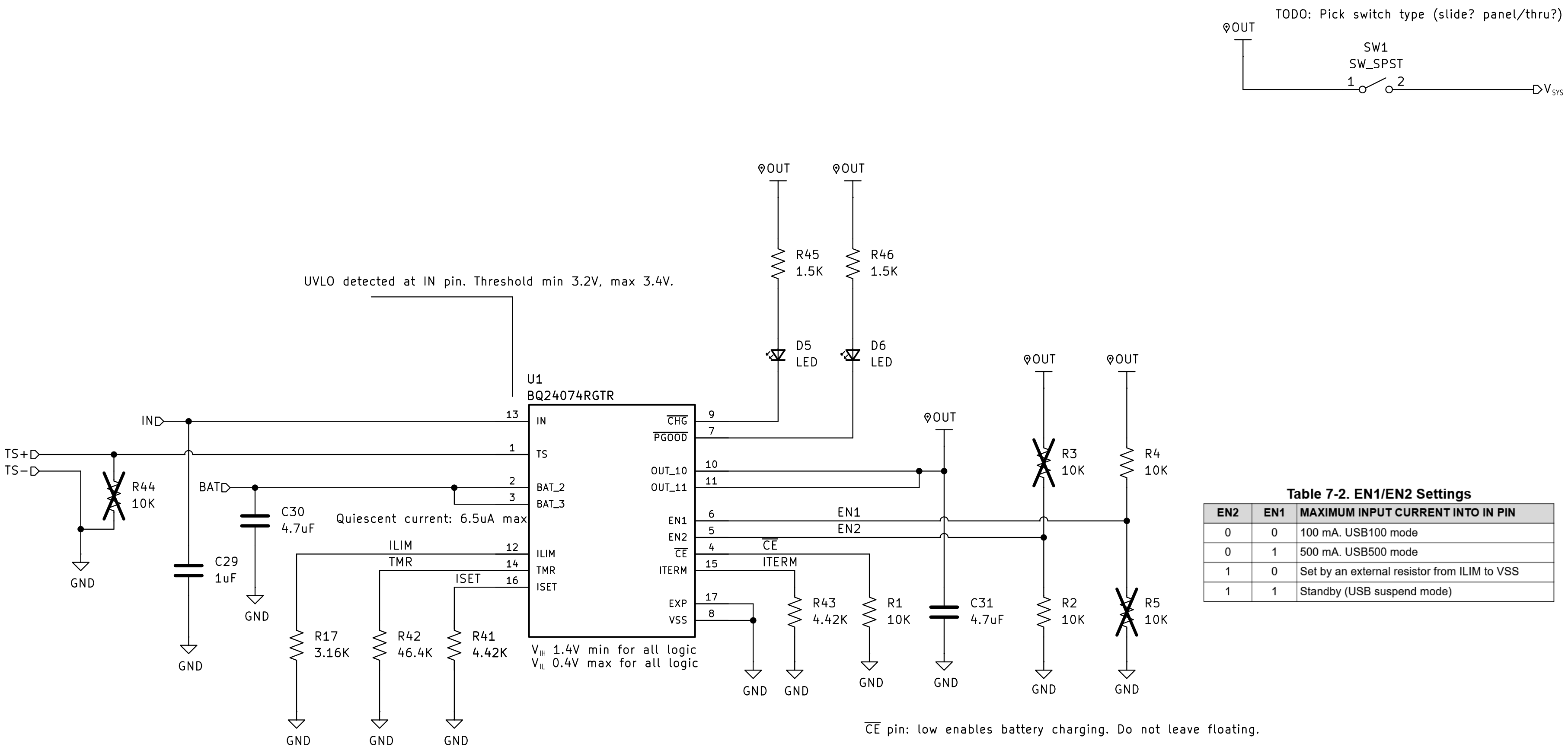


Table 7-2. EN1/EN2 Settings

EN2	EN1	MAXIMUM INPUT CURRENT INTO IN PIN
0	0	100 mA. USB100 mode
0	1	500 mA. USB500 mode
1	0	Set by an external resistor from ILIM to VSS
1	1	Standby (USB suspend mode)

$I_{LMAX} = K_{ILIM} / R_{ILIM}$
K_{ILIM} = 1550 A*Omhs
=> I_{LMAX} = 1500 A*Ohms / 3.16K = 475 mA

$I_{CHG} = K_{ISET} / R_{ISET}$
K_{ISET} = 890 A*Omhs
=> I_{CHG} = 890 A*Ohms / 4.42K = 201 mA

$t_{MAXCHG} = R_{TMR} * 10 * K_{TMR}$
K_{TMR} = 48 s*kOhms
=> t_{MAXCHG} = 46.4K * 48 s*kOhms * 10 = 6.2 hours

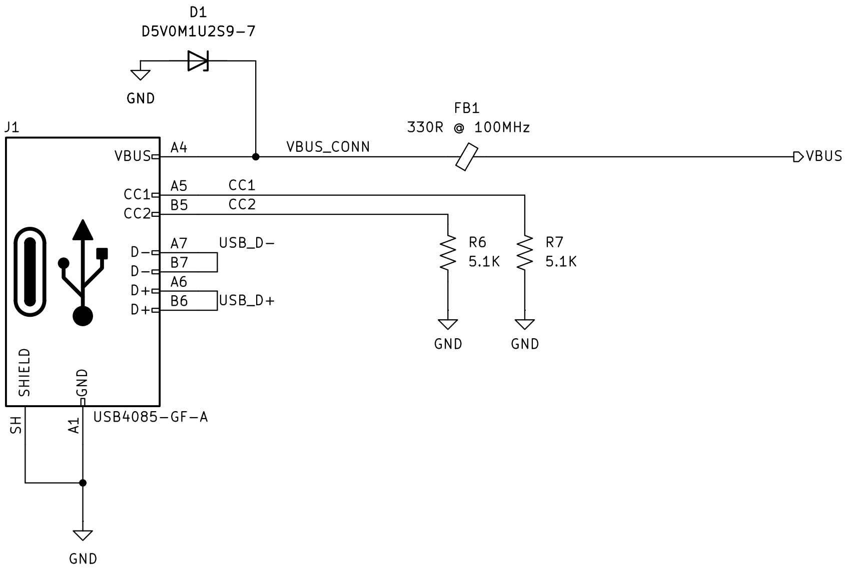
$I_{TERM} = 0.030 * R_{ITERM}/R_{ISET}$
R_{ISET} = 4.42K
=> I_{TERM} = 0.030 * 4.42K/4.42K = 30 mA

004−0001

Sheet: /Battery Charger/
File: charger.kicad_sch

Title: Portable Headphone Amp

Size: A4 | Date: 2026−08−25 | **Rev: A1**
KiCad E.D.A. 10.0.0 | Id: 2/9



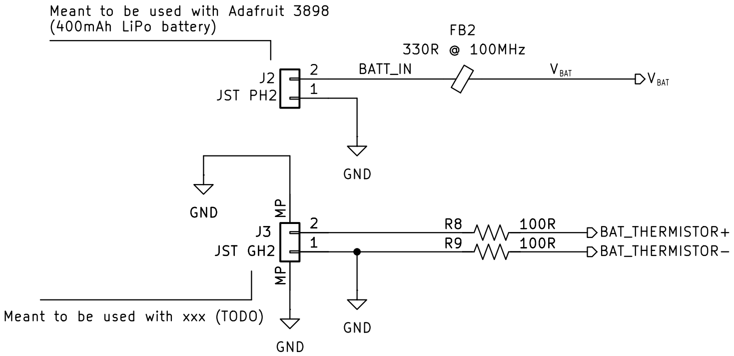
004−0001

Sheet: /Power Input/
File: power_input.kicad_sch

Title: Portable Headphone Amp

Size: A4 | Date: 2026−08−25 | **Rev: A1**

KiCad E.D.A. 10.0.0 | Id: 3/9

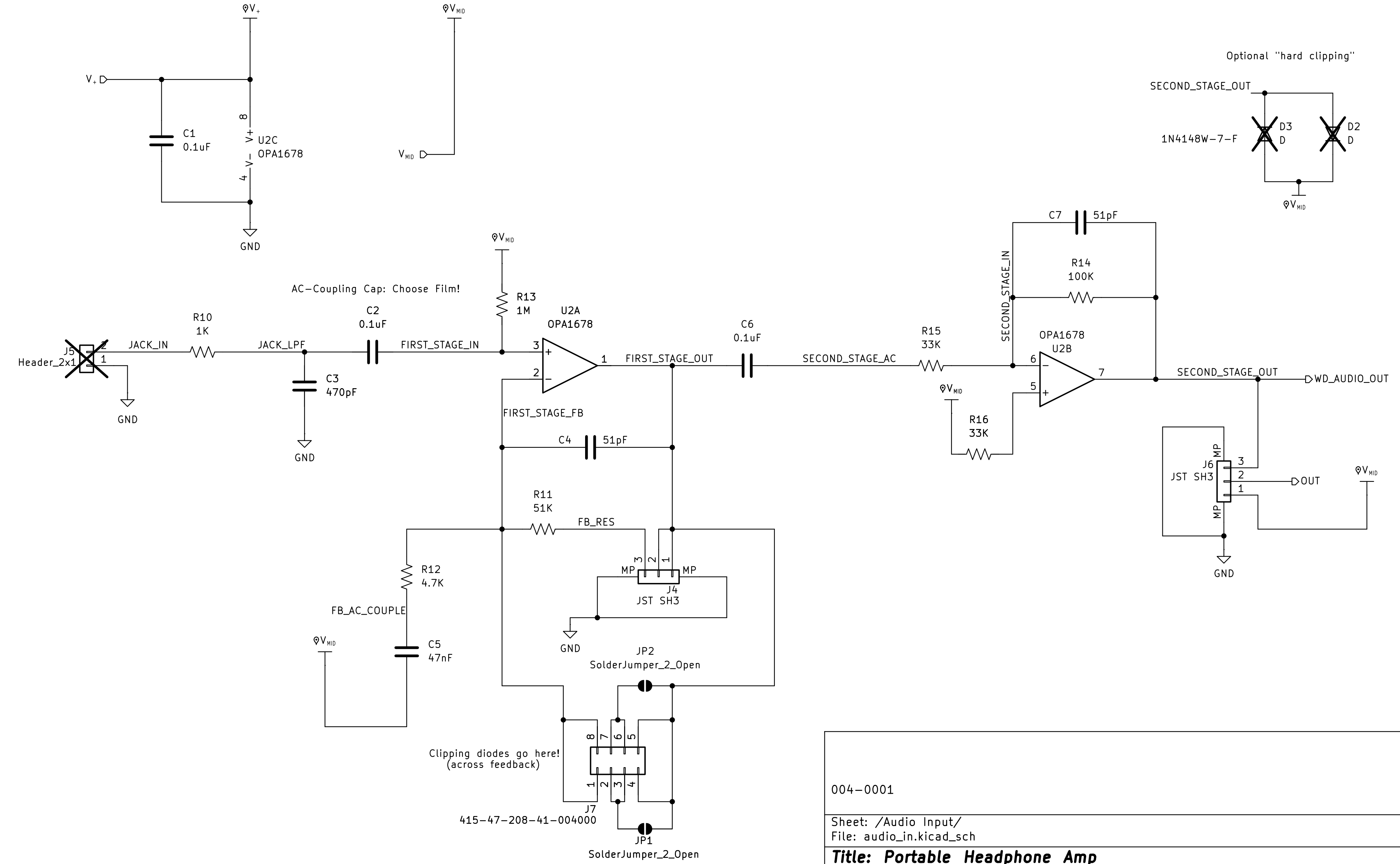


004−0001

Sheet: /Battery/
File: battery.kicad_sch

Title: Portable Headphone Amp

Size: A4	Date: 2026-08-25	**Rev: A1**
KiCad E.D.A. 10.0.0		Id: 4/9

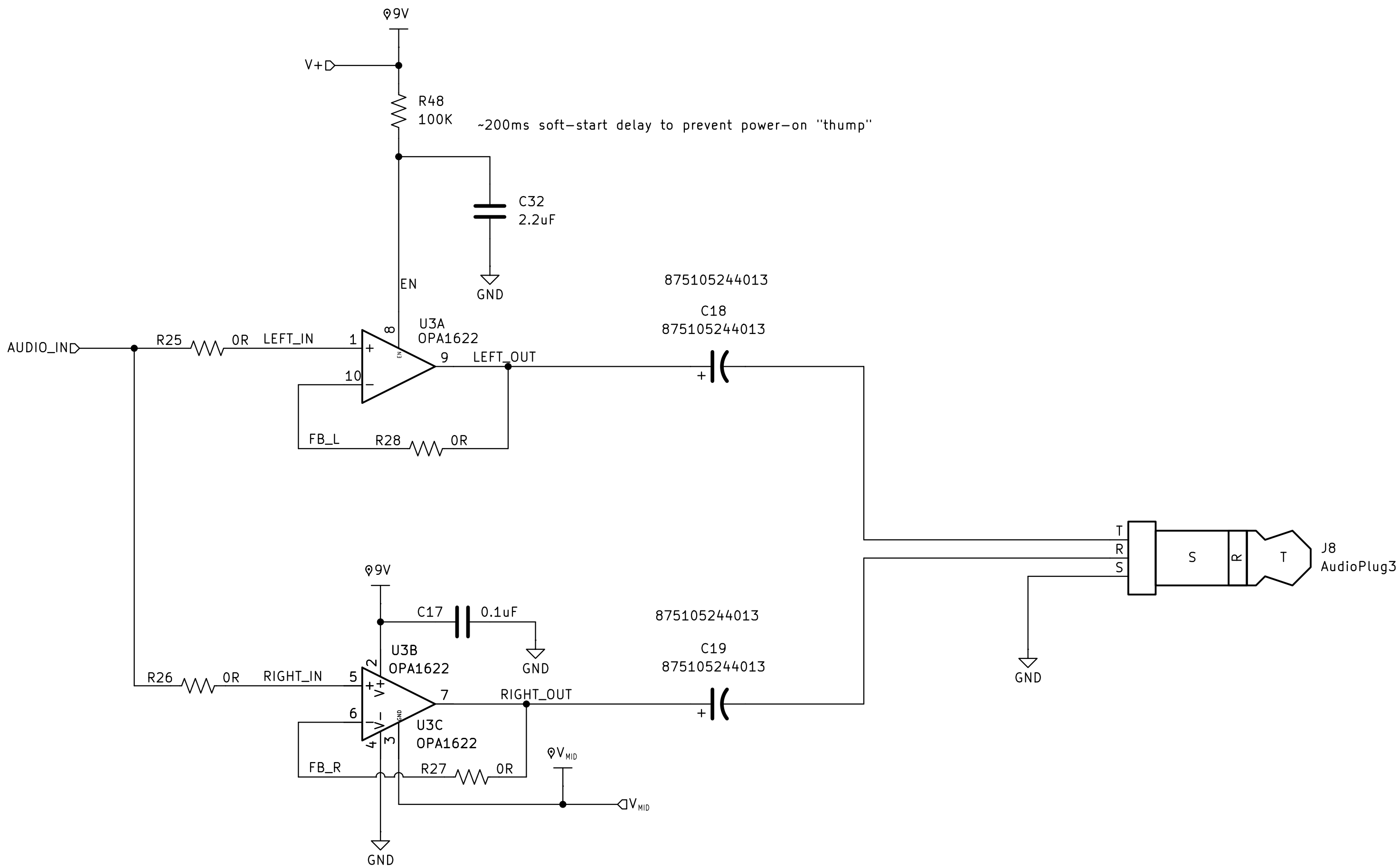


004-0001

Sheet: /Audio Input/
File: audio_in.kicad_sch

Title: Portable Headphone Amp

Size: A4 Date: 2026-08-25 **Rev: A1**
KiCad E.D.A. 10.0.0 Id: 5/9



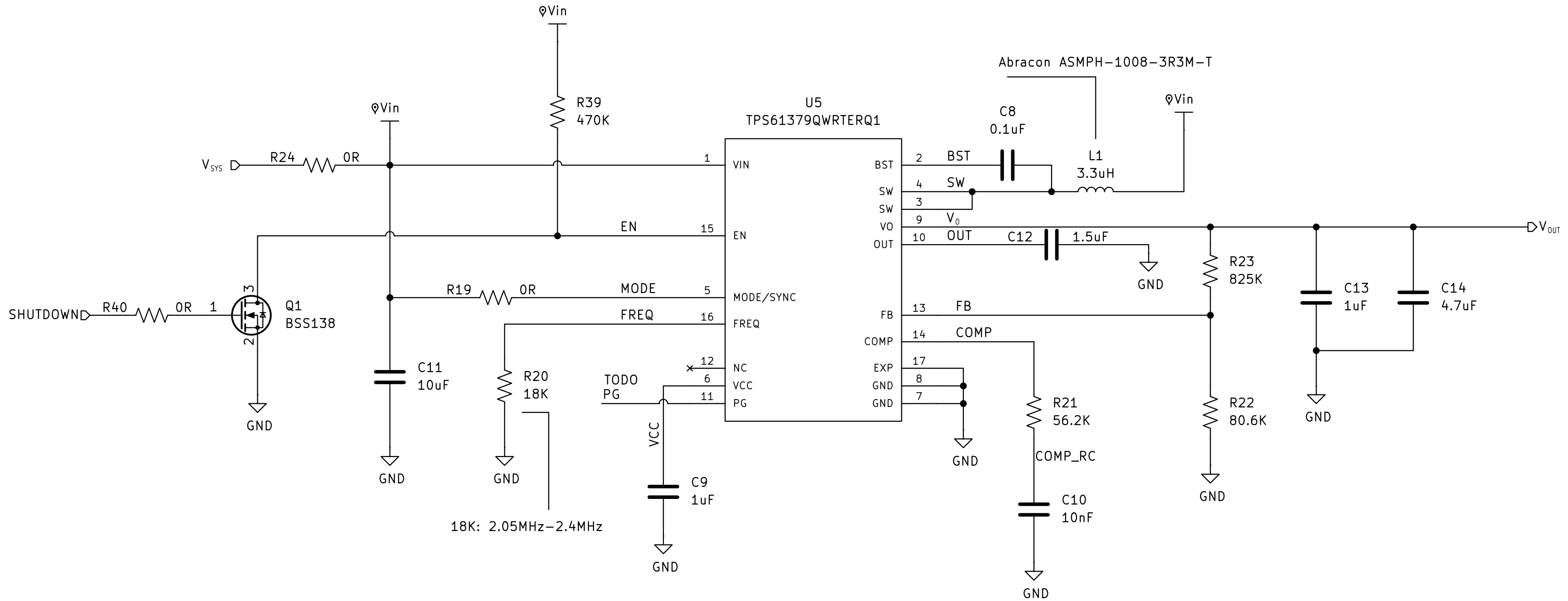
004-0001

Sheet: /Current Buffer/
File: buffer.kicad_sch

Title: Portable Headphone Amp

Size: A4	Date: 2026-08-25	Rev: A1
KiCad E.D.A. 10.0.0		Id: 6/9

WEBENCH Design (PDF Link)
9V, 0.2A out

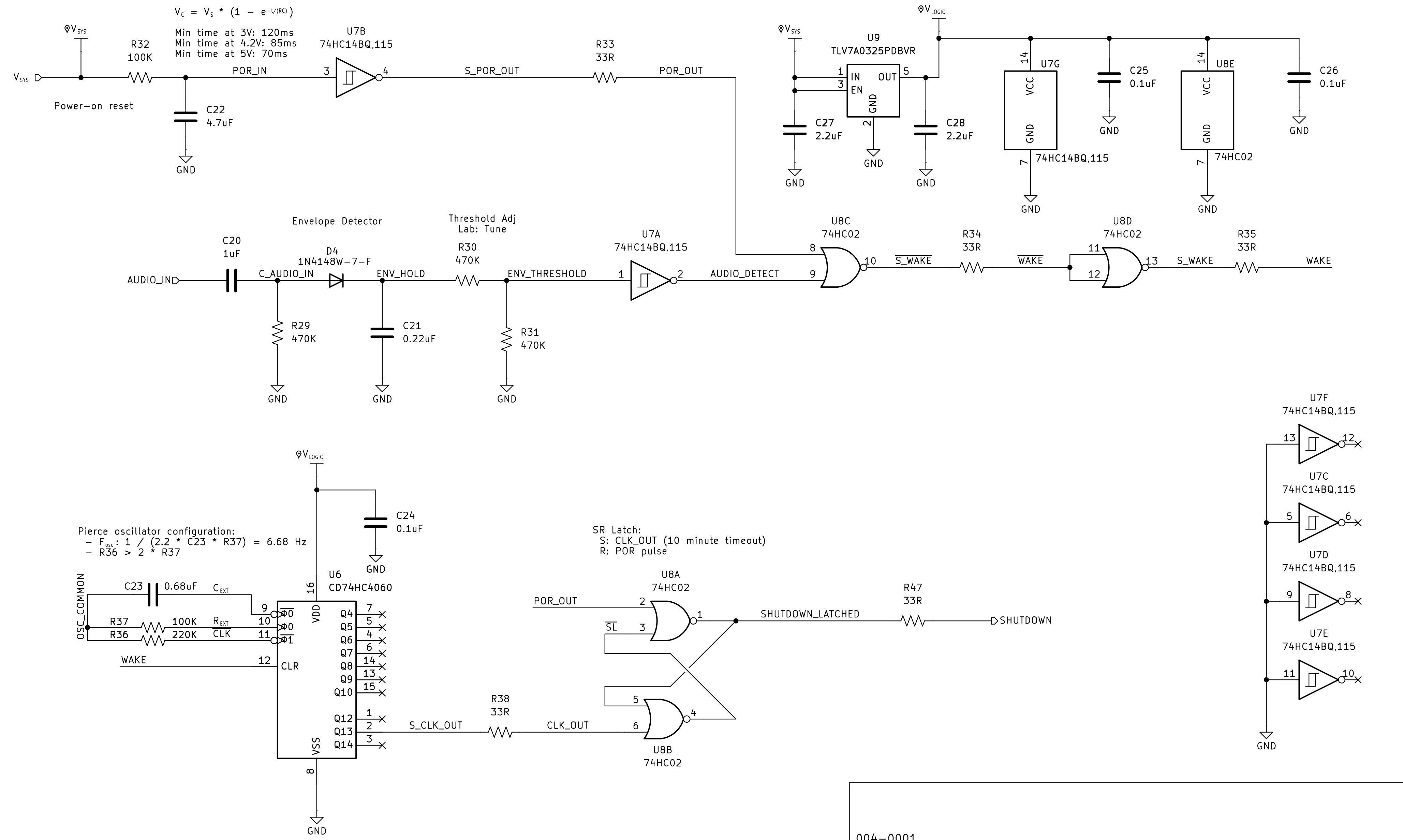


004–0001

Sheet: /Boost/
File: boost.kicad_sch

Title: Portable Headphone Amp

Size: A4	Date: 2026–08–25	**Rev: A1**
KiCad E.D.A. 10.0.0		Id: 7/9

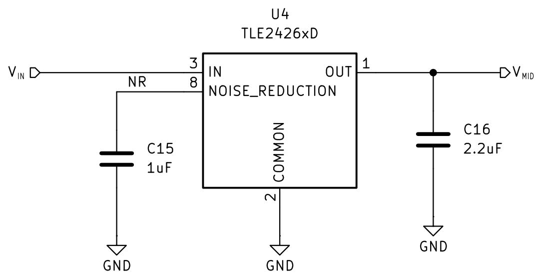


004-0001

Sheet: /Watchdog/
File: watchdog.kicad_sch

Title: Portable Headphone Amp

Size: A4	Date: 2026-08-25	**Rev: A1**
KiCad E.D.A. 10.0.0		Id: 8/9



004-0001

Sheet: /Rail Splitter/
File: rail_splitter.kicad_sch

Title: Portable Headphone Amp

Size: A4	Date: 2026-08-25	**Rev: A1**
KiCad E.D.A. 10.0.0		Id: 9/9